



Smart resource allocation and volunteer coordination for social impact.



Solution Challenge

A single operational view for intake, assignment, response tracking, and impact logging.

Intake

Allocation

Impact

Vercel

MVP preview

Administrative and user workflows are demonstrated through the prototype.

The deck includes proof screenshots, flow diagrams, and a deployment-ready layout.

Deployment direction: Vercel-hosted web prototype with centralized backend state.

RahatSetu

H2S
HACKERSKILL



Role-based
UI

Shared
state

Fast
routing

Team details

Prototype submission fields and project identity

Submission fields

Team name: ZeroLatency

Team leader name: Siddharth Chillapwar

Problem statement: Smart resource allocation and volunteer coordination for social impact.

Project identity

Project name: RahatSetu

Target users: citizens, volunteers, donors, and NGO admins

Scope: intake, assignment, analytics, impact tracking, and escalation handling.

Operational focus

NGO response portal and request routing.

Coordination layer

Volunteer dispatch and status visibility.

Outcome layer

Impact analytics and closure tracking.

What RahatSetu does

A concise operational system for NGO intake and response

Overview

RahatSetu unifies request intake, volunteer coordination, emergency reporting, donation handling, and NGO operations in one workflow.

Each submission is converted into a trackable case, so updates remain visible across all roles without duplicate records or fragmented follow-up.

Core promise

Intake → allocation → action → closure

Single source of truth

Real-time coordination

Feasible student prototype

1

Capture

2

Route

3

Resolve

4

Log impact

Opportunities

How the solution differs, solves, and scales

Why this is different

Relief workflows are usually split across forms, chat threads, spreadsheets, and local registers.

How it solves the problem

Every case moves through intake, assignment, response, and closure inside one shared workflow.

USP of the solution

Live heatmap + aggregate data + smart allocation + impact desk in one portal.

Opportunity statement

Build a single operational desk for NGO work instead of isolated tools.

Less duplication

Faster response

Cleaner tracking

Measurable impact

List of features offered by the solution

Separated by user role for clearer presentation

User-side features

- Login and dashboard home
- Resource desk
- Grievances and suggestions
- Emergency report submission
- Profile and notifications

Admin-side features

- Admin overview dashboard
- Aggregate data tracking
- Heatmap by location
- Smart allocation form
- Impact desk for results

Role-based UI

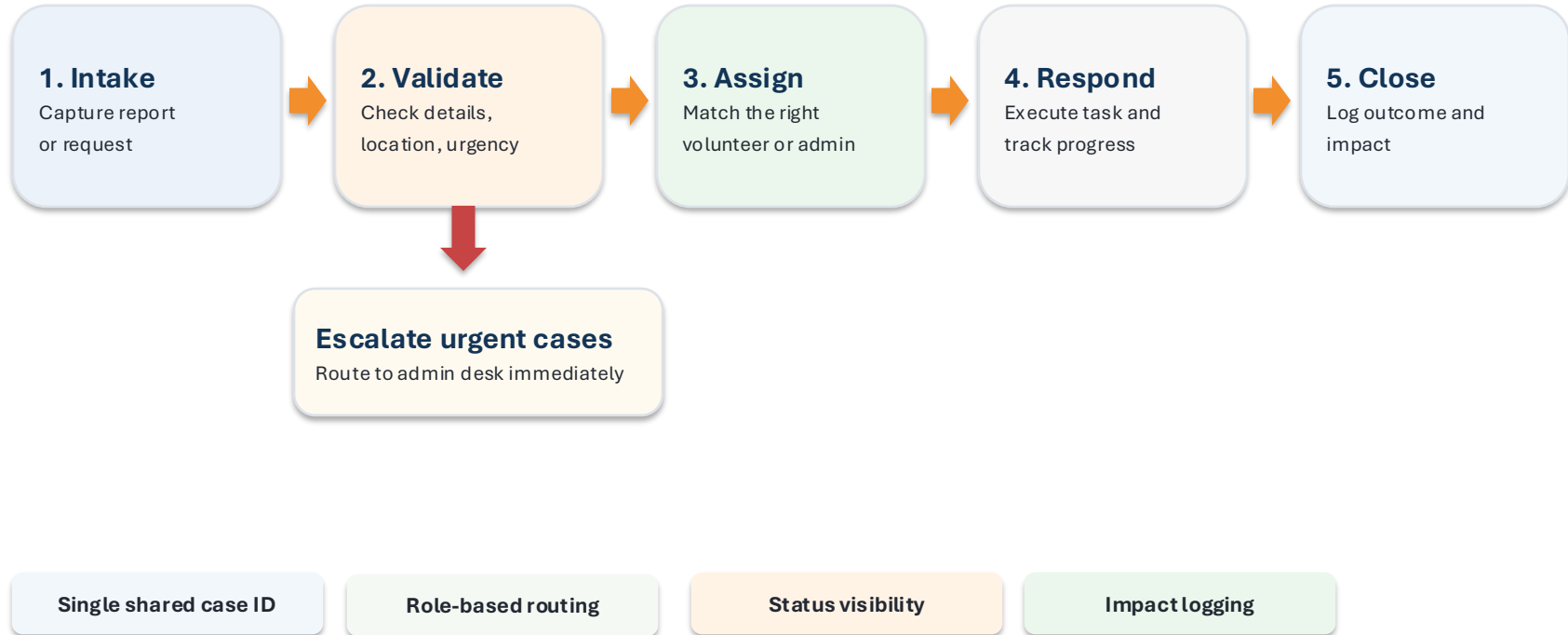
Heatmap

Alerts

Impact logging

Process flow diagram or Use-case diagram

A clearer operational sequence with one escalation branch

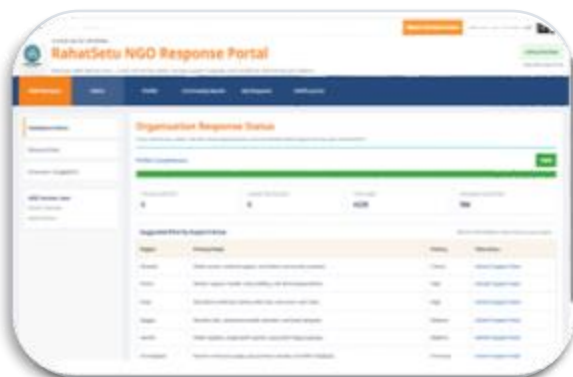


Wireframes / proof snapshots

Key screens from the prototype used as visual evidence



Login



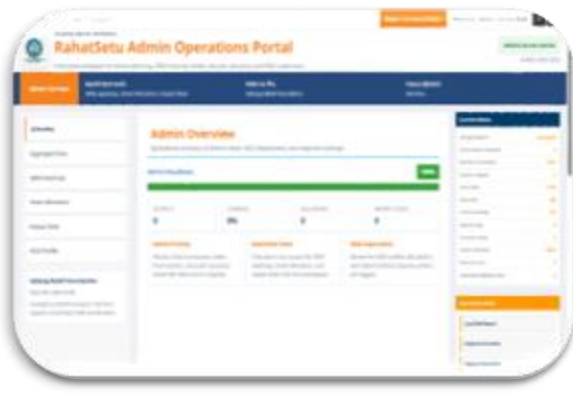
Dashboard home



Resource desk



Emergency report



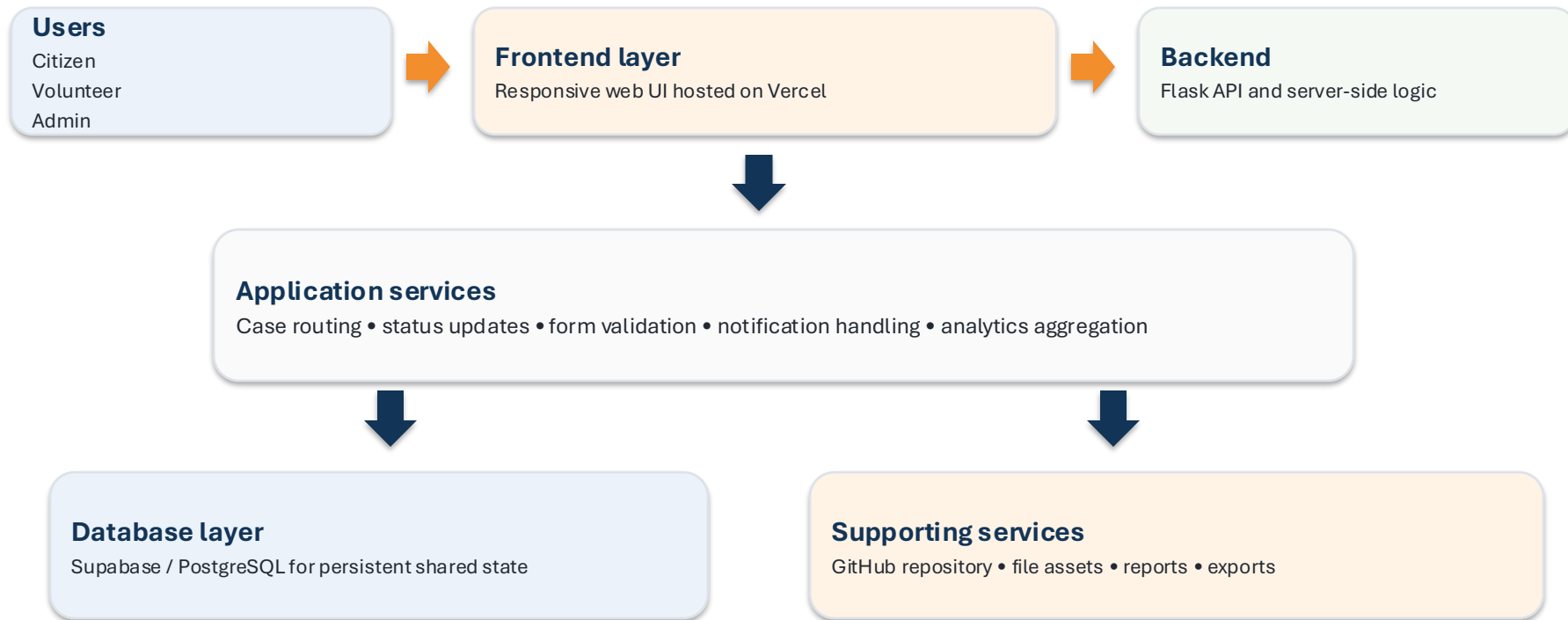
Admin overview



Heatmap proof

Architecture diagram of the proposed solution

Layered layout without redundant service boxes



Technologies to be used in the solution

Deployment, backend, data, and integration stack

Flask

Backend API and action handlers

HTML / CSS / JS

Responsive front end and dashboard UI

Supabase / Postgres

Persistent database and shared state

Vercel

Deployment platform for the web prototype

Antigravity

Integration and rapid prototyping support

Stitch

UI composition and asset stitching

GitHub collaboration

Deployment on Vercel

Role-based workflow

Fast iteration

Estimated implementation cost and links

Budget estimate, future scope, and external references

Estimated implementation cost

MVP can run on low-cost or free-tier infrastructure during development.

Primary scaling costs: compute, database growth, and maintenance.

Additional details / future development

Add automated volunteer matching, multilingual support, better auth, and notifications.

Extend analytics to SLA trends and downloadable reports.

Project links

GitHub Public Repository: <https://github.com/MADOUT20/GDG---SOLUTION-CHALLENGE->

MVP Link: <https://gdg-solution-challenge-jade.vercel.app/>

Working Prototype Link: <https://www.chaosfaction.xyz/>